A Novel Sliced Fused Gromov-Wasserstein Distance

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Abstract

The Gromov–Wasserstein (GW) distance and its fused extension (FGW) are powerful tools for comparing heterogeneous data. Their computation is, however, challenging since both distances are based on non-convex, quadratic optimal transport (OT) problems. Leveraging 1D OT, a sliced version of GW has been proposed to lower the computational burden. Unfortunately, this sliced version is restricted to Euclidean geometry and loses invariance to isometries, strongly limiting its application in practice. To overcome these issues, we propose a novel slicing technique for GW as well as for FGW that is based on an appropriate lower bound, hierarchical OT, and suitable quadrature rules for the underlying 1D OT problems. Our novel sliced FGW significantly reduces the numerical effort while remaining invariant to isometric transformations and allowing the comparison of arbitrary geometries. We show that our new distance actually defines a pseudo-metric for structured spaces that bounds FGW from below and study its interpolation properties between sliced Wasserstein and GW. Since we avoid the underlying quadratic program, our sliced distance is numerically more robust and reliable than the original GW and FGW distance; especially in the context of shape retrieval and graph isomorphism testing.

1 Introduction

The Gromov–Wasserstein (GW) distance (Mémoli 2011) and the Fused Gromov–Wasserstein (FGW) distance (Vayer et al. 2020a) extend the classical optimal transport (OT) framework (Villani 2003) to the comparison of heterogenous data by modelling them as metric measure spaces (mm-spaces). While the resulting distances are powerful tools, their computation is costly and does not admit an exact solution. To accelerate the underlying non-convex, quadratic program, several algorithms have been proposed using regularization techniques (Peyré, Cuturi, and Solomon 2016; Peyré and Cuturi 2019), employing linearized distances (Beier, Beinert, and Steidl 2022; Nguyen and Tsuda 2023), quantizating the problem (Chowdhury, Miller, and Needham 2021) or using constrained optimization (Scetbon, Peyré, and Cuturi 2022; Scetbon et al. 2023). As a consequence, established solvers mainly compute upper bounds and numerical approximations of the actual GW and FGW distance. An alternative acceleration framework builds on 1D OT and random projections to define the so-called sliced GW distance (Vayer et al. 2019; Beinert, Heiss, and Steidl 2023). However, sliced GW is restricted to Euclidean geometry and lacks isometric invariance, limiting its practical use. As a remedy, we design a novel sliced GW distance fixing these issues and extend it to FGW. In particular, our new distance allows for an analytical computation outlined in Figure 1.

For our design, we turn towards the complementary literature on lower GW bounds (Mémoli 2011; Vayer et al. 2019, 2020b; Jin, Yu, and Zhang 2022). Since many of these rely on optimization heuristics, such as bi-convex relaxation (Vayer et al. 2020b) or sorting (Vayer et al. 2019; Beinert, Heiss, and Steidl 2023), and we are interested in exact computation, we turn our attention to the bounds in (Mémoli 2011). In particular, we are motivated by recent theoretical work proving their metricity on certain spaces (Mémoli and Needham 2022) and the practical effectiveness of such bounds in graph isomorphism testing (Weitkamp 2022; Weitkamp et al. 2024), where established GW solvers fail in practice. The aim behind graph isomorphism testing (Shervashidze et al. 2011; Grohe and Schweitzer 2020) is to verify or disprove whether two or more graphs are topologically equivalent or not.

In this paper, we focus on the strongest of Mémoli's GW bounds—the third lower bound (TLB)—and start by extending TLB to the fused case. TLB itself is defined as a hierarchical OT problem, i.e., an OT problem whose costs are given by another OT problem. Indeed, formulations of such kinds recently have attracted attention from the OT community (Dukler et al. 2019; Alvarez-Melis and Fusi 2020). Despite interesting theoretical properties (Delon and Desolneux 2020; Bonet, Vauthier, and Korba 2025), these problems are usually computationally challenging. Therefore, recent work has focused on extending sliced Wasserstein (SW) techniques to the hierarchical case for acceleration (Piening and Beinert 2025; Nguyen et al. 2025). Keeping this technique in mind, we tackle the computational burden of our new FGW lower bound by rewriting it as an Euclidean Wasserstein distance. To circumvent the limitation of slicing in high dimensions (Tanguy, Flamary, and Delon 2025), we propose to employ numerical quadrature schemes. All in all, we deduce an equivalent, but more efficient, lower bound. Apart from extending the literature on sliced hierarchical

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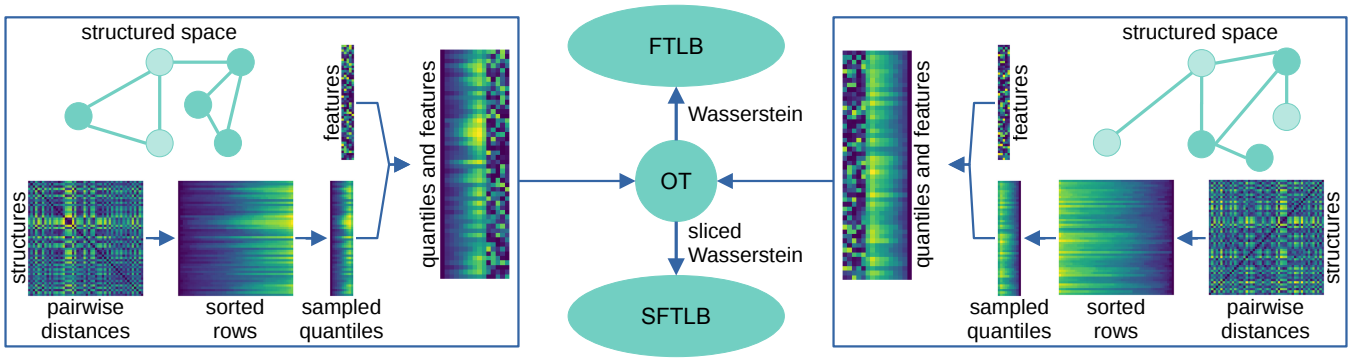


Figure 1: Visualization of our novel sliced fused Grovov–Wasserstein distance called SFTLB: Starting from two structured spaces like labeled graphs equipped with a graph distance, we sort the corresponding distance matrices row-wise. Subsequently, we determine samples of the quantile functions of the local distance distribution for each node. Afterward, we concatenate the sampled quantiles with the original features and interpret the outcome as an empirical Euclidean measure. Finally, we compute the Wasserstein or sliced Wasserstein distance between these measures.

OT, this also overcomes previous limitations of sliced GW, where sorting-based solvers might fail (Vayer et al. 2019; Beinert, Heiss, and Steidl 2023). We summarize our contributions as follows:

1. Towards the goal of an exact solver, we generalize Mémoli's GW bound to FGW and recall its connection to a classical Wasserstein problem.
2. To overcome the limitations of sliced GW, we employ this connection and numerical quadrature schemes to define an effective novel sliced FGW distance between arbitrary structured spaces beyond Euclidean geometry.
3. We prove that our new distance is a pseudo-metric interpolating between TLB and SW. Additionally, our proof illustrates the dimensional dependence of SW.
4. Lastly, we provide numerical experiments for our sliced FGW, showcasing its efficiency as well as its applicability for shape comparison and graph isomorphism testing.

Note that this is an extended version of a AAA'26 conference paper (Piening and Beinert 2026).

2 Optimal Transport-Based Distances

At the heart of OT is the calculation of optimal transfers from one measure to another. For this, let $\mathcal{P}(X)$ denote the space of Borel probability measures on a compact Polish space X. A measure $\xi \in \mathcal{P}(X)$ is translated to another compact Polish space Y via a mapping $T\colon X \to Y$ by the *push-forward* $T_\sharp\,\xi := \xi \circ T^{-1}$. Based on the canonical projections $\pi_\bullet$ onto a component, e.g., $\pi_X\colon X \times Y \to X, (x,y) \mapsto x$, the set of *transport plans* between $\xi \in \mathcal{P}(X)$, $\upsilon \in \mathcal{P}(Y)$ is defined as

$$\Gamma(\xi,\upsilon) := \big\{\gamma \in \mathcal{P}(X\times Y) \;\big|\; \pi_{X,\sharp}\,\gamma = \xi, \pi_{Y,\sharp}\,\gamma = \upsilon\big\}.$$

In general, OT seeks to find an optimal transport plan minimizing a given loss function. For certain losses, this procedure yields a metric between probability measures.

Wasserstein Distance

For compact $Z \subset \mathbb{R}^d$ equipped with the Euclidean norm $\|\cdot\|$, $\zeta, \zeta' \in \mathcal{P}(Z)$, and $p \in [1,\infty)$, the *p-Wasserstein distance* is introduced as

$$\mathrm{W}_p^p(\zeta,\zeta') := \inf_{\gamma\in\Gamma(\zeta,\zeta')} \underbrace{\int_{Z\times Z} \|z - z'\|^p \,\mathrm{d}\gamma(z,z')}_{=:\mathcal{T}_p(\gamma)} \tag{1}$$

and defines a metric[1] on $\mathcal{P}(Z)$. Its computation requires expensive optimization procedures since closed-form solutions are rarely available.

One exception is the special case $Z \subset \mathbb{R}$, where the optimal plan can be expressed using the *quantile function*:

$$q_\zeta\colon (0,1) \to \mathbb{R},\; s \mapsto \inf\big\{z \in \mathbb{R} \;\big|\; \zeta((-\infty,z]) > s\big\}.$$

Then, the p-Wasserstein distance admits the closed-form

$$\mathrm{W}_p^p(\zeta,\zeta') = \int_0^1 \big|q_\zeta(s) - q_{\zeta'}(s)\big|^p \,\mathrm{d}s; \tag{2}$$

see (Villani 2003). For discrete measures with with n and m support points, the integral in (2) reduces to a sum with $n+m-1$ summands; see (Peyré and Cuturi 2019, Prop. 3.5) and (Villani 2003, Thm. 2.18).

Sliced Wasserstein Distance

To leverage the computational benefits of the 1D Wasserstein distance, we consider the *slicing operator*

$$\pi_\theta\colon \mathbb{R}^d \to \mathbb{R},\; z \mapsto \theta \cdot z, \quad \theta \in \mathbb{S}^{d-1},$$

where $\mathbb{S}^{d-1} := \{\theta \in \mathbb{R}^d \mid \|\theta\| = 1\}$ denotes the sphere with surface area $\mathrm{A}(\mathbb{S}^{d-1})$ and $\cdot$ the Euclidean inner product. For compact $Z \subset \mathbb{R}^d$, $\zeta, \zeta' \in \mathcal{P}(Z)$, and $p \in [1,\infty)$, the *sliced p-Wasserstein* (SW) *distance* is introduced as

$$\mathrm{SW}_p^p(\zeta,\zeta') := \frac{1}{\mathrm{A}(\mathbb{S}^{d-1})} \int_{\mathbb{S}^{d-1}} \mathrm{W}_p^p(\pi_{\theta,\sharp}\,\zeta, \pi_{\theta,\sharp}\,\zeta')\,\mathrm{d}\theta$$

[1] Symmetric, positive definite function with triangle inequality.

with respect to the surface measure on $\mathbb{S}^{d-1}$. The interest in this sliced distance is justified by the close relation to the classical Wasserstein distances. Namely, for compact $Z \subset \mathbb{R}^d$, both distances display a form of metric equivalence, i.e., there exists $C_Z > 0$ such that

$$\mathrm{SW}_p^p(\zeta, \zeta') \leq \mathrm{W}_p^p(\zeta, \zeta') \leq C_Z \, \mathrm{SW}_p^{\frac{1}{(d+1)}}(\zeta, \zeta') \quad (3)$$

for any $\zeta, \zeta' \in \mathcal{P}(Z)$; see (Bonnotte 2013; Bonneel et al. 2015; Kolouri, Park, and Rohde 2016). Moreover, the spherical integral can be efficiently approximated via random projections using Monte Carlo or Quasi-Monte Carlo methods for low-dimensional data (Nguyen, Bariletto, and Ho 2024; Hertrich, Jahn, and Quellmalz 2025) as well as Gaussian approximations for high-dimensional data (Nadjahi, Vialard, and Peyré 2021). Note that there exist various extensions to non-Euclidean spaces (Kolouri et al. 2019; Quellmalz, Beinert, and Steidl 2023; Bonet, Drumetz, and Courty 2025), invariant data (Beckmann, Beinert, and Bresch 2025a,b), general probability divergences (Nadjahi et al. 2020; Hertrich et al. 2024) and kernels (Rux, Quellmalz, and Steidl 2025).

Gromov–Wasserstein Distance

To compare measures on distinct metric spaces, the GW distance extends classical OT by matching pairwise distances. For this, let $\mathbb{X} \coloneqq (X, g, \xi)$ be a *metric measure (mm) space* consisting of a compact metric space (X, g) and $\xi \in \mathcal{P}(X)$. For mm-spaces $\mathbb{X} \coloneqq (X, g, \xi)$ and $\mathbb{Y} = (Y, h, \upsilon)$, the *Gromov–Wasserstein* (GW) *distance* is then defined as

$$\mathrm{GW}_p^p(\mathbb{X}, \mathbb{Y}) \coloneqq \inf_{\gamma \in \Gamma(\xi, \upsilon)} \underbrace{\iint_{(X \times Y)^2} \big|g(x, x') - h(y, y')\big|^p \times \mathrm{d}\gamma(x', y') \, \mathrm{d}\gamma(x, y)}_{=: \mathcal{D}_p(\gamma)}. \quad (4)$$

Here, each transport plan π induces an overall distortion $\mathcal{D}_p$ between g and h. This expression defines a metric on the equivalence classes of isomorphic mm-spaces; see (Mémoli 2011). Since (4) is a non-convex, quadratic program, its minimization typically relies on entropic regularization and costly block-coordinate descent (Peyré and Cuturi 2019). As a consequence, a set of hierarchical lower bounds has been proposed in (Mémoli 2011; Mémoli and Needham 2022).

First Lower Bound Define the pointwise *p-eccentricity* $s_{\mathbb{X},p} \colon X \to \mathbb{R}$ of an mm-space $\mathbb{X} \coloneqq (X, g, \xi)$ as

$$s_{\mathbb{X},p}^p(x) \coloneqq \int_X g^p(x, x') \, \mathrm{d}\xi(x').$$

Then, the first lower GW bound between $\mathbb{X}$ and $\mathbb{Y}$ is

$$\mathrm{FLB}_p^p(\mathbb{X}, \mathbb{Y}) \coloneqq \mathrm{W}_p^p(s_{\mathbb{X},p,\sharp}\, \xi, s_{\mathbb{Y},p,\sharp}\, \upsilon),$$

which compares only the first-order moments of pairwise distances. This is an efficiently solvable 1D OT problem.

Second Lower Bound By matching the entire distribution of distances, the second lower GW bound reads as

$$\mathrm{SLB}_p^p(\mathbb{X}, \mathbb{Y}) \coloneqq \mathrm{W}_p^p\big(g_\sharp(\xi \otimes \xi), h_\sharp(\upsilon \otimes \upsilon)\big),$$

where $\otimes$ denotes the product measure. This lifts FLB by considering all pairwise distances at once. Again, characterization as 1D transport problem enables fast computation.

Third Lower Bound For each $(x, y) \in X \times Y$, the local distance distributions between $\mathbb{X}$ and $\mathbb{Y}$ is defined via

$$\mathrm{LD}_p^p(x, y) \coloneqq \inf_{\gamma \in \Gamma(\xi, \upsilon)} \int_{X \times Y} \big|g(x, x') - h(y, y')\big|^p \, \mathrm{d}\gamma(x', y') = \mathrm{W}_p^p\big(g(x, \cdot)_\sharp\, \xi, h(y, \cdot)_\sharp\, \upsilon\big). \quad (5)$$

Decoupling the minimization of the inner and outer integral in (4), we arrive at the third lower GW bound:

$$\mathrm{TLB}_p^p(\mathbb{X}, \mathbb{Y}) \coloneqq \inf_{\gamma \in \Gamma(\xi, \upsilon)} \underbrace{\int_{X \times Y} \mathrm{LD}_p^p(x, y) \, \mathrm{d}\gamma(x, y)}_{=: \bar{\mathcal{D}}_p(\gamma)}. \quad (6)$$

Note that the inner plans in (5) do not match the outer plan in (6). Similar to the other two cases, the estimation of LD benefits from accelerated 1D solutions. In contrast, TLB still requires a costly OT minimization.

Comparing all three lower bounds, we obtain the hierarchy (Mémoli 2011; Chowdhury and Mémoli 2019):

$$\mathrm{FLB}_p(\mathbb{X},\mathbb{Y}) \leq \mathrm{SLB}_p(\mathbb{X},\mathbb{Y}) \leq \mathrm{TLB}_p(\mathbb{X},\mathbb{Y}) \leq \mathrm{GW}_p(\mathbb{X},\mathbb{Y}).$$

Moreover, TLB is positive definite and thus defines a metric for certain classes of mm-spaces, whereas FLB and SLB fail to distinguish isomorphic mm-spaces (Mémoli and Needham 2022). To avoid the outer OT problem in (6), (Sato et al. 2020) rely on the maximum mean discrepancy (MMD) with respect to the 1D Wasserstein distance as kernel, which leads to an efficient computation scheme. In this paper, we follow a different approach by accelerating the computation of the outer OT problem using an efficient slicing, which actually leads to a metric equivalence with TLB.

Fused Gromov–Wasserstein Distance

The GW distance allows for a geometrically meaningful comparison between unlabeled, undirected graphs. However, due to the importance of node features in practical applications, practitioners often resort to a combination of Wasserstein and GW distances. For this, the mm-spaces are extended by a common compact feature space $Z \subset \mathbb{R}^d$. More precisely, a *structured space* $\mathbb{X}^{\mathcal{S}} \coloneqq (X \times Z, g, \xi)$ consists of a compact metric space (X, g) and $\xi \in \mathcal{P}(X \times Z)$. Structured spaces are also known as labelled spaces or structured objects (Vayer et al. 2020a). The space of all structured spaces is denoted as $\mathcal{S}(Z)$. Having the generalization of TLB and its slicing in mind, we study the following distance on $\mathcal{S}(Z)$, which is a specific variant of (Vayer et al. 2020a, Def. 8): for $\mathbb{X}^{\mathcal{S}} \coloneqq (X \times Z, g, \xi)$, and $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y \times Z', h, \upsilon)$ with $Z = Z'$ as well as $\alpha \in [0, 1]$ and $p \in [1, \infty)$, the *fused GW* (FGW) *distance* reads as

$$\mathrm{FGW}_{\alpha,p}^p(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \coloneqq \inf_{\gamma \in \Gamma(\xi, \upsilon)} (1 - \alpha)\, \mathcal{D}_p(\pi_{X \times Y, \sharp}\, \gamma) + \alpha\, \mathcal{T}_p(\pi_{Z \times Z', \sharp}\, \gamma),$$

where $\mathcal{D}_p$ denotes the distortion in (4), $\mathcal{T}_p$ refers to the transport in (1), and α balances structure versus features. In the case $\alpha = 0$, we recover the original GW distance of the structure part; for $\alpha = 1$, we obtain the original Wasserstein distance of the label part. Notice that, for $\alpha \in (0, 1)$ and

$p \in [1,\infty)$, FGW only defines a semi-metric[2] on the space of isomorphic structure spaces, fulfilling the relaxed triangle inequality

$$\begin{aligned}&\mathrm{FGW}^p_{\alpha,p}(\mathbb{Y}_1^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}})\\&\le 2^{p-1}\big(\mathrm{FGW}^p_{\alpha,p}(\mathbb{Y}_1^{\mathcal{S}},\mathbb{X}^{\mathcal{S}})+\mathrm{FGW}^p_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}})\big);\end{aligned}$$

see (Vayer et al. 2020a, Prop. 4).

3 Accelerating the Calculation of FGW

A Lower Bound Using TLB

In principle, any of the lower bounds for the GW distance may be directly extended to the FGW distance. Due to the metric properties of TLB and its tight affinity to the original GW distance, we solely focus on this bound. The basic idea is to replace the GW distortion in FGW by TLB. In this manner, for $\mathbb{X}^{\mathcal{S}} \coloneqq (X\times Z, g, \xi)$, $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y\times Z', h, \upsilon)$ with compact $Z = Z' \subset \mathbb{R}^d$, $\alpha \in [0,1]$, and $p \in [1,\infty)$, we introduce the *Fused Third Lower Bound* (FTLB) via

$$\begin{aligned}\mathrm{FTLB}^p_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) \coloneqq \inf_{\gamma\in\Gamma(\xi,\upsilon)} &(1-\alpha)\,\tilde{\mathcal{D}}_p(\pi_{X\times Y,\sharp}\,\gamma)\\ &+\alpha\,\mathcal{T}_p(\pi_{Z\times Z',\sharp}\,\gamma),\end{aligned} \tag{7}$$

where $\tilde{\mathcal{D}}_p$ relates to the mm-spaces $\mathbb{X} \coloneqq (X, g, \pi_{X,\sharp}\,\xi)$ and $\mathbb{Y} \coloneqq (Y, h, \pi_{Y,\sharp}\,\upsilon)$. Similar to GW and FGW, this includes TLB for $\alpha = 0$. Consequently, all subsequent results apply to the classical GW distance as well. Indeed, FTLB provides a lower bound since we minimize over multiple, uncoupled inner plans as opposed to finding a single coupled plan. Due to the similarity to FGW, many properties of the original distance carry over to FTLB.

Proposition 3.1. *For $\mathbb{X}^{\mathcal{S}} \coloneqq (X\times Z, g, \xi)$, $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y\times Z, h, \upsilon)$, $Z \subset \mathbb{R}^d$ compact, let $\xi^{\mathcal{F}} \coloneqq \pi_{Z,\sharp}\,\xi$, $\upsilon^{\mathcal{F}} \coloneqq \pi_{Z,\sharp}\,\upsilon$ and $\mathbb{X} \coloneqq (X, g, \pi_{X,\sharp}\,\xi)$, $\mathbb{Y} \coloneqq (Y, h, \pi_{Y,\sharp}\,\upsilon)$.*

(i) *There exists $\gamma^* \in \Gamma(\xi,\upsilon)$ minimizing* (7).
(ii) $\mathrm{FTLB}_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) \le \mathrm{FGW}_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}})$.
(iii) $\mathrm{FTLB}_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) \to \mathrm{W}_p(\xi^{\mathcal{F}},\upsilon^{\mathcal{F}})$ *as* $\alpha\to 1$.
(iv) $\mathrm{FTLB}_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) \to \mathrm{TLB}_p(\mathbb{X},\mathbb{Y})$ *as* $\alpha\to 0$.
(v) $\mathrm{FTLB}_{\alpha,p}$ *defines a pseudo-semi-metric*[3] *on $\mathcal{S}(Z)$ with*

$$\begin{aligned}&\mathrm{FTLB}^p_{\alpha,p}(\mathbb{Y}_1^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}})\\&\le 2^{p-1}\big(\mathrm{FTLB}^p_{\alpha,p}(\mathbb{Y}_1^{\mathcal{S}},\mathbb{X}^{\mathcal{S}})+\mathrm{FTLB}^p_{\alpha,p}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}})\big)\end{aligned}$$

for all $\mathbb{X}^{\mathcal{S}},\mathbb{Y}_1^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}} \in \mathcal{S}(Z)$.

A Novel Lower Bound via Slicing

While lowering the computational burden, FTLB still requires solving an OT problem. For two discrete measures of cardinality n, this leads to a computational complexity of $\mathcal{O}(n^3)$ for an exact OT solution and $\mathcal{O}(n^2\log n)$ for a regularized one (Peyré and Cuturi 2019). To achieve further acceleration, we aim to combine ideas from SW and numerical integration. For this, let $\mathbb{X}^{\mathcal{S}} \coloneqq (X\times Z, g, \xi)$ and $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y\times Z', h, \upsilon)$ be equipped with discrete measures

$$\xi \coloneqq \sum_{i=1}^n \xi_i\,\delta_{(x_i,z_i)} \quad\text{and}\quad \upsilon \coloneqq \sum_{j=1}^m \upsilon_j\,\delta_{(y_j,z'_j)}. \tag{8}$$

For the special case $p=2$, we start with considering LD_2 in (5), which is the key ingredient of the structure distortion $\tilde{\mathcal{D}}_2$. The required 1D Wasserstein distance may here be analytically determined by exploiting (2), where the quantile functions can be explicitly computed using efficient sorting algorithms. To surmount the outer OT problem of FTLB, we instead propose to merely approximate (2) via some quadrature scheme, i.e.,

$$\mathrm{LD}_2^2(x,y) \approx \sum_{k=1}^r w_k\,\big|q_{g(x,\cdot)\sharp\xi}(s_k) - q_{h(y,\cdot)\sharp\upsilon}(s_k)\big|^2 \tag{9}$$

with weights $w_k > 0$, knots $s_k \in (0,1)$, and small $r \in \mathbb{N}$. Note that we restrict ourselves to quadrature rules with positive weights. Using $\mathbf{q}_{\xi,x} \coloneqq (\sqrt{w}_k q_{g(x,\cdot)\sharp\xi}(s_k))_{k=1}^r$ and $\mathbf{q}_{\upsilon,y} \coloneqq (\sqrt{w}_k q_{h(y,\cdot)\sharp\upsilon}(s_k))_{k=1}^r$, we may also write

$$\mathrm{LD}_2^2(x,y) \approx \|\mathbf{q}_{\xi,x} - \mathbf{q}_{\upsilon,y}\|^2.$$

In this manner, FTLB can be approximated by

$$\begin{aligned}\mathrm{FTLB}^2_{\alpha,2}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) &\approx \inf_{\gamma\in\Gamma(\xi,\upsilon)} \sum_{i,j=1}^{n,m}\gamma_{ij}\left\|\begin{smallmatrix}\sqrt{1-\alpha}\,(\mathbf{q}_{\xi,x_i}-\mathbf{q}_{\upsilon,y_j})\\ \sqrt{\alpha}\,(z_i-z'_j)\end{smallmatrix}\right\|^2\\ &= \mathrm{W}_2^2\Big(\underbrace{\sum_{i=1}^n \xi_i\,\delta_{\left(\begin{smallmatrix}\sqrt{1-\alpha}\mathbf{q}_{\xi,x_i}\\ \sqrt{\alpha}z_i\end{smallmatrix}\right)}}_{=:\xi^{\mathcal{Q}}_\alpha\in\mathcal{P}(\mathbb{R}^{r+d})}, \underbrace{\sum_{j=1}^m \upsilon_j\,\delta_{\left(\begin{smallmatrix}\sqrt{1-\alpha}\mathbf{q}_{\upsilon,y_j}\\ \sqrt{\alpha}z'_j\end{smallmatrix}\right)}}_{=:\upsilon^{\mathcal{Q}}_\alpha\in\mathcal{P}(\mathbb{R}^{r+d})}\Big),\end{aligned} \tag{10}$$

where $\pi_{ij} \coloneqq \pi((x_i,z_i),(y_j,z'_j))$. This approximation carries over to TLB. More precisely, for $\mathbb{X} \coloneqq (X,g,\xi)$ and $\mathbb{Y} \coloneqq (Y,h,\upsilon)$ with $\xi \coloneqq \sum_{i=1}^n \xi_i\,\delta_{x_i}$ and $\upsilon \coloneqq \sum_{j=1}^m \upsilon_j\,\delta_{y_j}$, we here obtain

$$\mathrm{TLB}_2^2(\mathbb{X},\mathbb{Y}) \approx \mathrm{W}_2^2\Big(\underbrace{\sum_{i=1}^n \xi_i\,\delta_{\mathbf{q}_{\xi,x_i}}}_{=:\xi^{\mathcal{Q}}\in\mathcal{P}(\mathbb{R}^r)}, \underbrace{\sum_{j=1}^m \upsilon_j\,\delta_{\mathbf{q}_{\upsilon,y_j}}}_{=:\upsilon^{\mathcal{Q}}\in\mathcal{P}(\mathbb{R}^r)}\Big). \tag{11}$$

Up to here, we have approximated TLB and FTLB using a quadrature rule for the inner 1D Wasserstein distances. To overcome the remaining outer OT minimization behind the Wasserstein distances in (10) and (11), we finally propose to replace these with the SW distance:

$$\begin{aligned}\mathrm{SFTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) &\coloneqq \mathrm{SW}_2(\xi^{\mathcal{Q}}_\alpha,\upsilon^{\mathcal{Q}}_\alpha)\\ \mathrm{STLB}_2(\mathbb{X},\mathbb{Y}) &\coloneqq \mathrm{SW}_2(\xi^{\mathcal{Q}},\upsilon^{\mathcal{Q}}).\end{aligned} \tag{12}$$

In general, the sliced lower bounds hold only approximately, i.e., $\mathrm{SFTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) \lessapprox \mathrm{FGW}(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}})$, where the quality depends on the employed quadrature rule. Since $\xi^{\mathcal{Q}}_\alpha,\upsilon^{\mathcal{Q}}_\alpha \in \mathcal{P}(\mathbb{R}^{r+d})$ and $\xi^{\mathcal{Q}},\upsilon^{\mathcal{Q}} \in \mathcal{P}(\mathbb{R}^r)$, the employed SW distances can be efficiently computed independently of the atom numbers n and m in (8). Moreover, many properties of the original FGW distance from Proposition 3.1 carry over to the sliced variant.

[2] Metric without triangle inequality.
[3] Semi-metric d without definiteness ($d(x,x')=0 \not\Rightarrow x=x'$).

Proposition 3.2. *For $\mathbb{X}^{\mathcal{S}} \coloneqq (X \times Z, g, \xi)$, $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y \times Z, h, \upsilon)$, $Z \subset \mathbb{R}^d$ compact, let $\xi^{\mathcal{F}} \coloneqq \pi_{Z,\sharp}\,\xi$, $\upsilon^{\mathcal{F}} \coloneqq \pi_{Z,\sharp}\,\upsilon$ and $\mathbb{X} \coloneqq (X, g, \pi_{X,\sharp}\,\xi)$, $\mathbb{Y} \coloneqq (Y, h, \pi_{Y,\sharp}\,\upsilon)$. For $k, \ell \in \mathbb{N}$ with $\ell > 1$, let*

$$c_{k,\ell} \coloneqq \sqrt{\frac{\mathrm{A}(\mathbb{S}^{k+\ell+1})}{\mathrm{A}(\mathbb{S}^{\ell+1})} \frac{\mathrm{A}(\mathbb{S}^{\ell-1})}{\mathrm{A}(\mathbb{S}^{k+\ell-1})}}.$$

For an arbitrary quadrature rule, it holds:

(i) $\mathrm{SFTLB}_{\alpha,2}$ *defines a pseudo-metric* $\mathcal{S}(Z)$,
(ii) $\mathrm{SFTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \to c_{r,d}\,\mathrm{SW}_2(\xi^{\mathcal{F}}, \upsilon^{\mathcal{F}})$ *as* $\alpha \to 1$,
(iii) $\mathrm{SFTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \to c_{d,r}\,\mathrm{STLB}_2(\mathbb{X}, \mathbb{Y})$ *as* $\alpha \to 0$.

Given that $c_{k,\ell} \to 0$ for $k \to \infty$ and ℓ fixed (Gray 2004), the constants $c_{r,d}$ and $c_{d,r}$ illustrate the curse of dimensionality of the SW distance. In contrast to classical Wasserstein transport, the value of the SW distance for measures living on a subspace depends on the dimension of the underlying Euclidean space. Indeed, as we believe this to be of independent interest to the OT community, we state this dependence as a general result employed in the proof of Proposition 3.2.

Proposition 3.3. *For $\zeta, \zeta' \in \mathcal{P}(A \times Z)$, $A \subset \mathbb{R}^r$ and $Z \subset \mathbb{R}^d$ compact, with $\pi_{A,\sharp}\,\zeta = \pi_{A,\sharp}\,\zeta' = \delta_0$, it holds*

$$\mathrm{SW}_2(\zeta, \zeta') = c_{r,d}\,\mathrm{SW}_2(\pi_{Z,\sharp}\,\zeta, \pi_{Z,\sharp}\,\zeta').$$

Importantly, for empirical measures ξ and υ with $n = m$ and $\xi_i = \upsilon_i = 1/n$ in (8), the rather simple, equispaced midpoint rule with $r = n$ becomes exact, i.e.,

$$\mathrm{FTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) = \mathrm{W}_2(\xi_\alpha^{\mathcal{Q}}, \upsilon_\alpha^{\mathcal{Q}}) \tag{13}$$

instead of the approximation in (10). Consequently, SFTLB becomes an actual lower bound for the FGW distance. Moreover, SFTLB and FTLB are metrically equivalent.

Proposition 3.4. *For $\mathbb{X}^{\mathcal{S}} \coloneqq (X \times Z, g, \xi)$ and $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y \times Z, h, \upsilon)$ with bounded X and Y, i.e. $g(x, x') \leq D$ and $h(y, y') \leq D$ for fixed D, and compact $Z \subset \mathbb{R}^d$, let ξ and υ in* (8) *be empirical measures with $n = m$ and $\xi_i = \upsilon_i = 1/n$. For the equispaced midpoint rule with $r = n$, it holds:*

(i) $\mathrm{SFTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \leq \mathrm{FGW}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}})$,
(ii) *there exists $C_{Z,D,r} > 0$ such that*

$$\mathrm{SFTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \leq \mathrm{FTLB}_{\alpha,2}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}})$$
$$\leq C_{Z,D,r}\,\mathrm{SFTLB}_{\alpha,2}^{\frac{1}{r+d+1}}(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}). \tag{14}$$

Note that $C_{Z,D,r}$ can be chosen such that (14) holds true for all $\mathbb{X}^{\mathcal{S}}$ and $\mathbb{Y}^{\mathcal{S}}$ of the stated form. Furthermore, Proposition 3.4 can be directly generalized to empirical measures ξ and υ with different atom numbers n and m by using a non-equispaced midpoint rule with $r = n + m - 1$, see the remark in our technical appendix.

4 Numerical Experiments

In the following, we study the computational runtime of our distances, the underlying transport plans, as well as their behaviour in applications like free-support barycenters, shape classification, and graph isomorphism testing. All experiments[4] were conducted with 13th Gen Intel Core i5-13600K CPU and an NVIDIA GeForce RTX 3060 GPU (12 GB).

[4]Code: github.com/MoePien/slicing_fused_gromov_wasserstein

n	100	1000	10 000
FGW	0.03 ± 0.02	0.45 ± 0.07	178.29 ± 5.22
FTLB	0.06 ± 0.03	0.10 ± 0.02	15.69 ± 0.43
SFTLB	0.04 ± 0.04	0.03 ± 0.01	1.18 ± 0.00

Table 1: Average computation time in seconds for 5 instances with graph sizes 100, 1000, and 10 000. The remaining parameters have been chosen as $r = 10$ and $L = 50$.

Computational and Runtime Analysis

For simplicity, we assume that the measures of the structured spaces $\mathbb{X}^{\mathcal{S}} \coloneqq (X \times Z, g, \xi)$ and $\mathbb{Y}^{\mathcal{S}} \coloneqq (Y \times Z, h, \upsilon)$ are empirical with the same number of atoms, i.e., $n = m$ and $\xi_i = \upsilon_j = 1/n$ in (8). The first step of FTLB and SFTLB is to sample the quantiles of the local distance distributions, i.e., to compute $\mathbf{q}_{\xi,x_i}$ and $\mathbf{q}_{\upsilon,y_j}$ in (11), which essentially requires a row-wise sorting of the distance matrices regarding g and h with total complexity $\mathcal{O}(n^2 \log n)$.

FTLB In its second step, FTLB requires solving the outer OT problem in (11). Calculating the OT costs requires $\mathcal{O}((r + d)\, n^2)$ operations. The OT problem itself may be solved exactly in $\mathcal{O}(n^3)$ or approximately in $\mathcal{O}(n^2 \log n)$ using entropic regularization; see (Peyré and Cuturi 2019). This leads to an all-in-all complexity of $\mathcal{O}((r + d + n)\, n^2)$ (exact) or $\mathcal{O}((r+d+\log n)\, n^2)$ (approximated). In the experiments, we perform exact inner OT calculations with $r = n$.

SFTLB Here, we initially determine $\xi_\alpha^{\mathcal{Q}}$ and $\upsilon_\alpha^{\mathcal{Q}}$. Next, we employ a Monte Carlo scheme to calculate (12) by

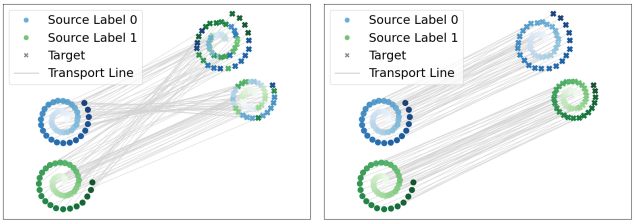
$$\mathrm{SFTLB}_{\alpha,2}^2(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \approx \frac{1}{L} \sum_{\ell=1}^{L} \mathrm{W}_2^2(\pi_{\theta_\ell,\sharp}\,\xi_\alpha^{\mathcal{Q}}, \pi_{\theta_\ell,\sharp}\,\upsilon_\alpha^{\mathcal{Q}})$$

with uniformly distributed directions $\theta_\ell \in \mathbb{S}^{r+d-1}$. The 1D Wasserstein distance can be computed by sorting, yielding an all-in-all complexity of $\mathcal{O}(Ln(r+d) + (Ln + n^2)\log n)$, where we aim for a small r. Here, the Monte Carlo scheme converges in $\mathcal{O}(\sqrt{L})$ (Nadjahi et al. 2020).

The computational complexity of FTLB and SFTLB is dominated by the preliminary row-wise sorting and the employed OT solvers. To study the actual run-time speed-up, we generate 5 pairs of $n \times n$ distance matrices with $n \in \{100, 1000, 10\,000\}$ and 1D normal features. Using the POT toolbox (Flamary et al. 2021) for the OT computations, we then determine the corresponding FTLB, SFTLB, and FGW distance. The runtimes are recorded in Table 1, where the significant speed-up due to the slicing is clearly visible.

FTLB Transport Plans

Although the GW distance is well fitted to detect similarities between two single structures, it usually fails to detect these in a set of independent structures. The FGW distance can overcome this issue by assigning appropriate labels. This observation carries over to the outer transport plans of TLB and FTLB, which is illustrated in Figure 2, where the left two spirals are transported to the right two. Source and target are here modelled as point clouds in $\mathbb{R}^2$ equipped with the Euclidean distance and binary feature labels.



(a) TLB (b) FTLB ($\alpha = 0.9$)

Figure 2: Comparison of TLB and FTLB transport plans. Crosses are colored based on the transported mass. FTLB achieves a more regular transport plan by integrating labels.

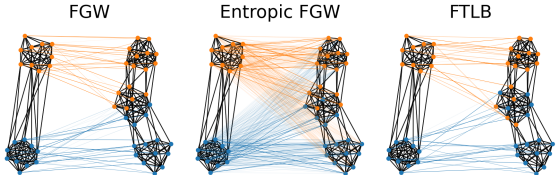


Figure 3: Comparison of FGW, entropic FGW ($\epsilon = 0.01$), FTLP transport plan for $\alpha = 0.5$. Target nodes are colored according to the transported mass. FGW and FTLB capture the same structural similarities.

To compare our FTLB transport with FGW and its entropic version, we adapt an example from (Flamary et al. 2021); see Figure 3. Each image displays the source graph on the left and the target graph on the right. The transports are visualized using the color of the community assignment in the source. In this example, the FTLB plan captures the main structures similarly to the original FGW plan, indicating that our FTLB transport may be an efficently-to-compute alternative in practice.

Free-Support Euclidean Barycenters

GW barycenters are generalized Fréchet means that interpolate several mm-spaces $\mathbb{Y}_k$, $k = 1, \dots, K$ by solving

$$\arg\min \left\{ \sum_{k=1}^{K} \mathrm{GW}_2^2(\mathbb{X}, \mathbb{Y}_k) \;\middle|\; \mathbb{X} \text{ mm-space} \right\}. \tag{15}$$

The solutions of the GW barycenter problem can be completely characterized using multi-marginal OT (Beier, Beinert, and Steidl 2023; Beier and Beinert 2025); its numerical computation, however, remains challenging. Established free-support methods (Peyré and Cuturi 2019) minimize over the distance of $\mathbb{X}$ producing non-Euclidean outputs; whereas fixed-support methods (Beier, Beinert, and Steidl 2023; Beier et al. 2025) limit the resolution of the barycenter. As an alternative, we replace GW in (15) by TLB, STLB ($r = 500, L = 500$) and the so-called Anchor Energy (AE) distance (Sato et al. 2020) for comparison. For each distance, we perform a gradient descent over the points $x_i \in \mathbb{R}^d$ in $\mathbb{X} = (\{x_i\}_{i=1}^n, d_{\mathrm{E}}, \iota)$, where d_{E} is the Euclidean metric and ι the uniform measure. Figure 4 shows computed 2D barycenters for different upsampled shapes ($n = 500$) from

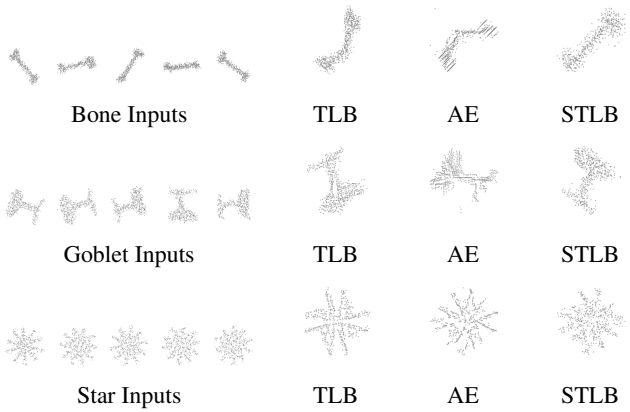


Figure 4: Euclidean 2D TLB, AE and STLB barycenters for five samples of different shape classes.

(Beier, Beinert, and Steidl 2022), obtained via three identical random restart initializations, and 1000 steps (width: 0.1).

Classification of 2D and 3D Shapes

A key application of the GW distance is the comparison of shapes, which can be represented as Euclidean point clouds or as meshes equipped with geodesic distances. Building on experiments from (Mémoli 2011; Beier, Beinert, and Steidl 2022), we rely on the k-nearest neighbor (KNN) classification (Cover and Hart 1967). More precisely, we first precompute pairwise distance matrices with respect to SLB, TLB, STLB ($r = 10, L = 100$), AE (Sato et al. 2020), qGW (Chowdhury and Mémoli 2019), LGW (Beier, Beinert, and Steidl 2022) and GW. Second, we evaluate the classification accuracy by assigning each test point to the majority class among its three nearest neighbors. We report the mean accuracy over 1000 random 25%/75% training/test splits on the following datasets:

2D Shapes: 80 shapes from 4 classes (bone, goblet, star, horseshoe) represented as Euclidean point clouds, each consisting of 50 points (Beier, Beinert, and Steidl 2022);

Animals: 73 human/animal shapes from 7 classes (Sumner and Popović 2004) downsampled to 3D meshes with 50 vertices, equipped with the geodesic distance (Mémoli 2011; Beier, Beinert, and Steidl 2022);

FAUST-500 and FAUST-1000: 100 human meshes from 10 subjects (Bogo et al. 2014), downsampled and represented by 3D meshes with 500 and 1000 vertices equipped with the geodesic distance using PyVista (Sullivan and Kaszynski 2019) and NetworkX (Hagberg, Schult, and Swart 2008).

Table 2 summarizes classification accuracy, runtime, and nullspace types (Mémoli and Needham 2022): SLB vanishes for globally matching distance distributions ('global'), TLB/STLB/AE for locally matching distributions ('local'), and GW for isomorphic shapes ('iso'). Lastly, LGW and

distance	type	2D shapes		animals		FAUST-500		FAUST-1000	
		acc. (%)	time (ms)	acc. (%)	time (ms)	acc. (%)	time (ms)	acc. (%)	time (ms)
SLB	global	97.9±2.2	0.6	100.0±0.1	0.7	31.2±5.4	48.3	35.2 ± 5.7	183.7
TLB	local	100.0±0.3	0.7	100.0±0.0	0.8	36.8±5.4	24.8	40.0 ± 6.0	88.3
STLB	local	99.3±1.2	1.3	99.2±1.1	1.5	35.8±5.5	11.6	40.1±5.7	23.1
AE	local	99.7±0.9	0.6	97.8±1.8	0.7	37.8±5.6	8.1	41.6±5.7	35.8
GW	iso	99.7±0.6	2.6	100.0±0.0	5.8	29.2±4.3	482.0	33.3±5.2	2398.3
LGW	approx	91.2±2.7	288.9	89.8±3.9	481.0	29.4±4.7	$7 \cdot 10^4$	30.2±5.1	10^5
qGW	approx	61.6±4.4	16.4	48.8±6.3	14.7	13.7±3.6	18.2	12.4±3.2	35.3

Table 2: KNN Shape classification: Mean accuracy (acc.) with standard deviation and mean runtime (time).

graph model	WS	BA	RR	WS	BA	RR
nodes	10	10	10	500	500	100
parameters	k=4, p_E=0.1	m=5, p_B=0.5	$R=3$, $p_B=0.5$	k=250, p_E=0.1	m=250, p_B=0.1	$R=3$, $p_B=0.5$
SLB	90.40 ± 2.62	61.50 ± 3.44	96.30 ± 1.62	50.00 ± 0.00	50.00 ± 0.00	100.00 ± 0.00
TLB	96.80 ± 2.99	99.50 ± 1.02	99.40 ± 1.80	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
STLB	99.00 ± 1.00	99.80 ± 0.40	99.80 ± 0.60	99.60 ± 0.80	99.40 ± 0.92	100.00 ± 0.00
GW	67.40 ± 3.83	73.30 ± 3.85	65.50 ± 4.03	65.00 ± 5.00	100.00 ± 0.00	100.00 ± 0.00
WL-D	100.00 ± 0.00	100.00 ± 0.00	50.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	50.00 ± 0.00
FTLB	100.00 ± 0.00	96.00 ± 1.18	97.60 ± 1.50	100.00 ± 0.00	94.80 ± 1.60	98.60 ± 0.92
SFTLB	100.00 ± 0.00	98.20 ± 1.08	99.40 ± 0.92	100.00 ± 0.00	100.00 ± 0.00	99.80 ± 0.60
FGW	98.20 ± 1.66	90.30 ± 1.79	74.60 ± 2.69	100.00 ± 0.00	98.80 ± 1.33	69.80 ± 4.42
WL-F	100.00 ± 0.00	83.10 ± 3.21	81.30 ± 3.85	100.00 ± 0.00	99.20 ± 1.33	86.80 ± 3.92

Table 3: Mean accuracy (%) with standard deviation for graph isomorphism testing. The first five methods are label-agnostic, whereas the last four methods employ node features. Proposed methods are printed in bold.

qGW approximate GW ('approx')[5]. While most methods perform near-perfectly on '2D shapes' and 'animals', accuracy drops on FAUST datasets. On these, 'local' methods (TLB, STLB, AE) consistently outperform other approaches. Although GW should yield perfect classifications in theory, the underlying non-convex quadratic minimization reduces the numerical accuracy. STLB shows a clear runtime advantage for large-scale datasets.

Detection of Graph Isomorphisms

An important theoretical property of the GW and FGW distance is that they vanish only for two isomorphic inputs (Mémoli 2011; Vayer et al. 2020a), which facilitates an application in graph isomorphism testing. In practice, the established GW and FGW solvers are however not accurate enough to give reliable results of this NP-hard problem (Grohe and Schweitzer 2020). Therefore, SLB has been proposed as a statistical test for graph isomorphisms (Weitkamp 2022; Weitkamp et al. 2024). However, SLB has less statistical power than TLB (Mémoli and Needham 2022); for this reason, we propose to use STLB and SFTLB instead. During the experiment, we rely on synthetic datasets of random graphs with fixed node numbers, where half of the graph pairs are isomorphic and half of the graph pairs are not. Here, we consider these graph generation models:

Watts–Strogatz (WS) Graph: The WS graph is governed by a neighborhood number k and an edge rewiring probability p_E (Watts and Strogatz 1998). We add node features according to a 1D standard normal distribution.

[5]LGW/qGW depend heavily on hyperparameter choices balancing speed and performance, see our appendix for a discussion.

Barabasi–Albert (BA) Graph: The BA graph is governed by an edge generation number m (Barabási and Albert 1999). We assign node features in $\{0, 1\}$ based on a Bernoulli distribution with probability p_B.

Random Regular (RR) Graph: The RR graph is based on a regularity R (Bollobás 1998). We assign Bernoulli node features in $\{0, 1\}$ according to the probability p_B.

We compare the generated, unlabeled graph pairs using SLB, TLB, STLB, GW, and Weisfeiler-Lehman (WL) kernel test (Shervashidze et al. 2011). For the WL test, we use 5 iterations and set the labels to the node degree (WL-D). The labeled graph pairs are compared using $\alpha = 0.5$ for FTLB, SFTLB ($r = 5$, $L = 100$), and entropic FGW ($\varepsilon = 10^{-3}$), as well as with WL using 5 iterations and (binned) node features (WL-F). We classify two graphs as isomorphic if they are among the 50% most similar graph pairs in terms of the different distances or if the distance becomes exactly zero. In Table 3, we display the results of 10 experiment repetitions over 6 different random graph configurations.

5 Conclusion

We address the computational limits of GW and FGW by introducing new lower bounds. Extending TLB to the fused setting and linking it to a Wasserstein problem, we derive an efficient sliced FGW via numerical quadrature of 1D OTs. Our work bridges TLB and SW, clarifies SW's dimensional behavior, and proves effective for shape and graph comparisons, thereby overcoming the limitations of sliced GW (Vayer et al. 2019). In future research, the calculation of local distance distribution required for AE, FTLB, and SFTLB may be further accelerated via sweep-line algorithms (Sato et al. 2020), which is not studied in this work.

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A Technical Appendix

Detailed Proofs of the Propositions

Proof of Prop. 3.1. (i) Lebesgue's dominated convergence theorem implies $g(x_n,\cdot)_\sharp\,\xi \to g(x,\cdot)_\sharp\,\xi$ weakly if $x_n \to x$ in X and $h(y_n,\cdot)_\sharp\,\upsilon \to h(y,\cdot)_\sharp\,\upsilon$ weakly if $y_n \to y$ in Y. Thus, the local distance distribution LD_p is continuous on $X \times Y$. Since the Euclidean distance is continuous as well, the infimum in (7) is attained (Santambrogio 2015, Thm. 1.4).

(ii) Similarly to the relation between GW and TLB (Mémoli 2011, Prop. 6.3), for any $\gamma \in \Gamma(\pi_{X,\sharp}\,\xi, \pi_{Y,\sharp}\,\upsilon)$, we obtain

$$\begin{aligned}\mathcal{D}_p(\gamma) &= \iint_{(X\times Y)^2} |g(x,x') - h(y,y')|^p \,\mathrm{d}\gamma(x',y')\,\mathrm{d}\gamma(x,y)\\ &\geq \int_{X\times Y}\left[\inf_{\gamma'\in\Gamma(\pi_{X,\sharp}\,\xi,\pi_{Y,\sharp}\,\upsilon)} \int_{X\times Y} |g(x,x') - h(y,y')|^p \,\mathrm{d}\gamma'(x',y')\right]\mathrm{d}\gamma(x,y)\\ &\int_{X\times Y} \mathrm{LD}_p(x,y)\,\mathrm{d}\gamma(x,y) = \tilde{\mathcal{D}}_p(\gamma).\end{aligned}$$

(iii) From the definition of FTLB, part (ii), and the interpolation properties of FGW (Vayer et al. 2020a, Thm. 3), we obtain

$$\alpha\,\mathrm{W}_p^p(\pi_{Z,\sharp}\,\xi, \pi_{Z',\sharp}\,\upsilon) \leq \mathrm{FTLB}_{\alpha,p}^p(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \leq \mathrm{FGW}_{\alpha,p}^p(\mathbb{X}^{\mathcal{S}}, \mathbb{Y}^{\mathcal{S}}) \xrightarrow[\alpha\to 1]{} \mathrm{W}_p^p(\pi_{Z,\sharp}\,\xi, \pi_{Z',\sharp}\,\upsilon).$$

Since the left-hand side has the same limit as the right-hand side, the statement is established.

(iv) From the definition of FTLB and the compactness of Z, we deduce

$$(1-\alpha)\,\mathrm{TLB}_p^p(\mathbb{X},\mathbb{Y}) \leq \mathrm{FTLB}_{\alpha,p}^p(\mathbb{X}^{\mathcal{S}},\mathbb{Y}^{\mathcal{S}}) \leq (1-\alpha)\,\mathrm{TLB}_p^p(\mathbb{X},\mathbb{Y}) + \alpha\,C,$$

where C is an upper bound for the Wasserstein part, i.e., $\int_{Z\times Z}\|z-z'\|^p\,\mathrm{d}\eta \leq C$ for all $\eta \in \Gamma(\pi_{z,\sharp}\gamma_X, \pi_{z,\sharp}\gamma_Y)$. Here, the second inequality holds true, due to the gluing lemma (Villani 2003, Lem. 7.6). More precisely, gluing the optimal TLP plan in $\Gamma(\pi_{X,\sharp}\,\xi, \pi_{Y,\sharp}\,\upsilon)$ with ξ and υ, we obtain an FTLP plan in $\Pi(\xi,\upsilon)$. Taking the limit $\alpha \to 0$ establishes the statement.

(v) Positivity and symmetry are clear. For the triangle inequality, let $\gamma_i \in \mathcal{P}(X \times Y_i)$ be the optimal FTLB plan between $\mathbb{X}^{\mathcal{S}} \coloneqq (X \times Z, g, \xi)$ and $\mathbb{Y}_i^{\mathcal{S}} \coloneqq (Y \times Z_i, h_i, \upsilon_i)$ with $Z = Z_i$ and $i = 1,2$. By the gluing lemma (Villani 2003, Lem. 7.6), there exists $\tilde{\gamma} \in \mathcal{P}((X\times Z)\times(Y_1\times Z_1)\times(Y_2\times Z_2))$ with $\pi_{(X\times Z)\times(Y_i\times Z_i)\sharp}\,\tilde{\gamma} = \gamma_i$. Using the triangle inequality of the Wasserstein and Euclidean distances as well as the convexity of the power function, we obtain

$$\begin{aligned}\mathrm{FTLB}_{\alpha,p}^p(\mathbb{Y}_1^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}}) &\leq \int_{(X\times Z)\times(Y_1\times Z_1)\times(Y_2\times Z_2)} (1-\alpha)\,\mathrm{LD}_p^p(y_1,y_2) + \alpha\,\|z_1-z_2\|^p \;\mathrm{d}\tilde{\gamma}((x,z),(y_1,z_1),(y_2,z_2))\\ &\leq \int_{(X\times Z)\times(Y_1\times Z_1)\times(Y_2\times Z_2)} (1-\alpha)\big(\mathrm{LD}_p(x,y_1)+\mathrm{LD}_p(x,y_2)\big)^p + \alpha\big(\|z-z_1\|+\|z-z_2\|\big)^p\,\mathrm{d}\tilde{\gamma}((x,z),(y_1,z_1),(y_2,z_2))\\ &\leq 2^{p-1}\Big(\int_{(X\times Z)\times(Y_1\times Z_1)} (1-\alpha)\;\mathrm{LD}_p^p(x,y_1) + \alpha\,\|z-z_1\|^p\;\mathrm{d}\gamma_1((x,z),(y_1,z_1))\\ &\qquad + \int_{(X\times Z)\times(Y_2\times Z_2)} (1-\alpha)\;\mathrm{LD}_p^p(x,y_2) + \alpha\,\|z-z_2\|^p\;\mathrm{d}\gamma_2((x,z),(y_2,z_2))\Big)\\ &= 2^{p-1}\big(\mathrm{FTLB}_{\alpha,p}^p(\mathbb{X}^{\mathcal{S}},\mathbb{Y}_1^{\mathcal{S}}) + \mathrm{FTLB}_{\alpha,p}^p(\mathbb{X}^{\mathcal{S}},\mathbb{Y}_2^{\mathcal{S}})\big).\end{aligned}$$

□

Proof of Prop. 3.2. (i) Depending on the employed quadrature rule, the mapping $\xi \mapsto \xi_\alpha^{\mathcal{Q}}$ in (11) may not be injective; therefore, the SW distance in (12) only implies the pseudo-metric properties of SFTLB.

(ii) By construction, we have $\xi_\alpha^{\mathcal{Q}} \to \xi_1^{\mathcal{Q}}$ and $\upsilon_\alpha^{\mathcal{Q}} \to \upsilon_1^{\mathcal{Q}}$ weakly in $\mathcal{P}(\mathbb{R}^{r+d})$ as $\alpha \to 1$. Since SW is continuous with respect to the weak topology (Nadjahi et al. 2020, Thm. 1), it follows $\mathrm{SW}_2(\xi_\alpha^{\mathcal{Q}}, \upsilon_\alpha^{\mathcal{Q}}) \to \mathrm{SW}_2(\xi_1^{\mathcal{Q}}, \upsilon_1^{\mathcal{Q}})$. Observing that the support of $\xi_1^{\mathcal{Q}}$ and $\upsilon_1^{\mathcal{Q}}$ is contained in $\{0\}\times\mathbb{R}^d$, we deduce the assertion using Proposition 3.3.

(iii) By construction, we have $\xi_\alpha^{\mathcal{Q}} \to \xi_0^{\mathcal{Q}}$ and $\upsilon_\alpha^{\mathcal{Q}} \to \upsilon_0^{\mathcal{Q}}$ weakly in $\mathcal{P}(\mathbb{R}^{r+d})$ as $\alpha \to 0$. Since SW is continuous with respect to the weak topology (Nadjahi et al. 2020, Thm. 1), it follows $\mathrm{SW}_2(\xi_\alpha^{\mathcal{Q}}, \upsilon_\alpha^{\mathcal{Q}}) \to \mathrm{SW}_2(\xi_0^{\mathcal{Q}}, \upsilon_0^{\mathcal{Q}})$. Observing that the support of $\xi_0^{\mathcal{Q}}$ and $\upsilon_0^{\mathcal{Q}}$ is contained in $\mathbb{R}^r\times\{0\}$, we deduce the assertion using Proposition 3.3. □

Proof of Prop. 3.3. To establish the statement, we parametrize the integral of the SW distance by the first dimension. Following (Müller 1998, §1), every $\theta \in \mathbb{S}^{r+d-1}$ can be written as $(t, \sqrt{1-t^2}\,\theta')$ with $t \in [-1,1]$ and $\theta' \in \mathbb{S}^{r+d-2}$. For the integral, this means $\mathrm{d}\theta = (1-t^2)^{\frac{r+d-3}{2}}\,\mathrm{d}t\,\mathrm{d}\theta'$. Moreover, the considered measures ζ and ζ' are only supported on the last d components of $\mathbb{R}^{r+d}$ by assumption. Now we denote the projection to the last $r+d-1$ components by $P_1^\perp(x_1,\ldots,x_{r+d}) \coloneqq (x_2,\ldots,x_{r+d})$.

Since ζ and ζ' are supported on $0 \times \mathbb{R}^{r+d-1}$, the projection $P_1^\perp$ is actually a bijection on their supports. Therefore, the push-forwards $\tilde{\zeta} \coloneqq P_{1,\sharp}^\perp \zeta$ and $\tilde{\zeta}' \coloneqq P_{1,\sharp}^\perp \zeta'$ can be reverted, and we have $\zeta = \delta_0 \otimes \tilde{\zeta}$ and $\zeta' = \delta_0 \otimes \tilde{\zeta}'$. Incorporating the parametrization and the projection into the slicing, we obtain

$$\pi_{\theta,\sharp}\,\zeta = \big\langle \big(\tfrac{t}{\sqrt{1-t^2}\theta'} \big), \cdot \big\rangle_\sharp \, \zeta = \big(\sqrt{1-t^2}\, \cdot \big)_\sharp \langle \theta', \cdot \rangle_\sharp \, \tilde{\zeta} = \big(\sqrt{1-t^2}\, \cdot \big)_\sharp \pi_{\theta',\sharp}\, \tilde{\zeta}$$

Furthermore, we exploit that the Wasserstein distance is homogenous, i.e., $\mathrm{W}_2^2((\lambda\cdot)_\sharp \eta, (\lambda\cdot)_\sharp \eta') = \lambda^2\, \mathrm{W}_2^2(\eta, \eta')$ for all $\lambda \in \mathbb{R}$ and for all compactly supported η, η'; see (Villani 2003, Prop. 7.16.i). With these preliminary considerations, we deduce

$$\begin{aligned}
\mathrm{SW}_2^2(\zeta, \zeta') &= \frac{1}{\mathrm{A}(\mathbb{S}^{r+d-1})} \int_{\mathbb{S}^{r+d-1}} \mathrm{W}_2^2(\pi_{\theta,\sharp}\,\zeta, \pi_{\theta,\sharp}\,\zeta')\,\mathrm{d}\theta \\
&= \frac{1}{\mathrm{A}(\mathbb{S}^{r+d-1})} \int_{\mathbb{S}^{r+d-2}} \int_{-1}^{1} \mathrm{W}_2^2(\pi_{(t,\sqrt{1-t^2}\theta'),\sharp}\,\zeta, \pi_{(t,\sqrt{1-t^2}\theta'),\sharp}\,\zeta')\,(1-t^2)^{\frac{r+d-3}{2}}\,\mathrm{d}t\,\mathrm{d}\theta' \\
&= \frac{1}{\mathrm{A}(\mathbb{S}^{r+d-1})} \int_{\mathbb{S}^{r+d-2}} \int_{-1}^{1} \mathrm{W}_2^2(\pi_{\theta',\sharp}\,\tilde{\zeta}, \pi_{\theta',\sharp}\,\tilde{\zeta}')\,(1-t^2)^{\frac{r+d-1}{2}}\,\mathrm{d}t\,\mathrm{d}\theta' \\
&= \frac{\mathrm{A}(\mathbb{S}^{r+d+1})}{\mathrm{A}(\mathbb{S}^{r+d})} \frac{\mathrm{A}(\mathbb{S}^{r+d-2})}{\mathrm{A}(\mathbb{S}^{r+d-1})}\, \mathrm{SW}_2^2(\tilde{\zeta}, \tilde{\zeta}'),
\end{aligned}$$

where the integral with respect to t has been calculated in (Müller 1998, (§1.35)). Repeating this argumentation r times, we obtain the assertion. □

Proof of Prop. 3.4. (i) The statement directly follows from Proposition 3.1 in combination with (13) and (3).

(ii) The metric equivalence follows from the exactness in (13), the corresponding equivalence (3) of the Wasserstein and SW distance, and the definition of SFTLB in (12). □

Extension of Prop. 3.4. In order to extent Prop. 3.4 to the unbalanced case $n \neq m$, we have to choose a quadrature scheme such that (9) holds precisely instead of approximately, and thus (13) remains valid. For this, notice that the push-forwards $g(x,\cdot)_\sharp \xi$ and $h(y,\cdot)_\sharp \upsilon$ remain empirical measures and that the corresponding quantile functions $q_{g(x,\cdot)_\sharp \xi}$ and $q_{h(y,\cdot)_\sharp \upsilon}$ for all $x \in X$ and $y \in Y$ are step functions with exactly n and m equal length steps, respectively. The difference $q_{g(x,\cdot)_\sharp \xi} - q_{h(y,\cdot)_\sharp \upsilon}$ is also a step function, but with maybe unequal step lengths. The jumps in these differences, however, occur at the same positions for all $x \in X$ and $y \in Y$. Choosing the knots s_k in the middle of the steps and the weights w_k as the related step length, we obtain an exact quadrature scheme with $r = n + m - 1$ for (9). □

B Further Experiments

Additional Runtime Experiment

We extend the runtime experiment presented in Table 1, where we compared our method with the exact OT solver from POT. Again, we employ random Euclidean distance matrices based on randomly initialized point clouds. Here, we present additional runtime experiments of SFTLB compared to FTLB with entropic regularization (Peyré and Cuturi 2019). Since we observe slow POT runtimes for entropically regularized algorithms, we turn to another package. In particular, we employ the '*geomloss*' package (Feydy et al. 2019) for entropic OT with strong (0.1) and weak entropic regularization (0.001). Since the '*geomloss*' package is incapable of handling $10\,000 \times 10\,000$ matrices, we repeat the previous runtime experiment with graph sizes 50, 500, and 5 000. Additionally, we present the SFTLB runtime with different configurations of the quadrature size r and the projection number L. Again, we run our experiments on the CPU. Looking at the table, we again see a runtime advantage of SFTLB compared to FTLB. Additionally, we observe that L has a strong impact on the runtime, whereas the impact of r is more subtle.

n	50	500	5 000
Entropic FTLB (Regularization: 0.1)	0.00 ± 0.00	0.05 ± 0.01	2.95 ± 0.07
Entropic FTLB (Regularization: 0.001)	0.01 ± 0.01	0.06 ± 0.01	4.14 ± 0.17
SFTLB ($r = 10$ & $L = 50$)	0.00 ± 0.00	0.04 ± 0.01	0.30 ± 0.00
SFTLB ($r = 10$ & $L = 1000$)	0.00 ± 0.00	0.09 ± 0.01	0.70 ± 0.01
SFTLB ($r = 100$ & $L = 50$)	0.00 ± 0.00	0.05 ± 0.02	0.31 ± 0.02
SFTLB ($r = 100$ & $L = 1000$)	0.01 ± 0.00	0.07 ± 0.01	0.74 ± 0.01

Table 4: Average computation time in seconds for 5 instances with graph sizes 50, 500, and 5 000.

Illustration of Hyperparameter Impact

Next, we illustrate the impact of L and r visually. To illustrate the convergence of our Monte Carlo estimate, we plot

$$\frac{\frac{1}{L}\sum_{\ell=1}^{L} \mathrm{W}_2^2(\pi_{\theta_\ell,\sharp}\,\xi_\alpha^{\mathcal{Q}}, \pi_{\theta_\ell,\sharp}\,\upsilon_\alpha^{\mathcal{Q}})}{\frac{1}{20\,000}\sum_{\ell=1}^{20\,000} \mathrm{W}_2^2(\pi_{\theta_\ell,\sharp}\,\xi_\alpha^{\mathcal{Q}}, \pi_{\theta_\ell,\sharp}\,\upsilon_\alpha^{\mathcal{Q}})}$$

for different choices of L, 10 different random Euclidean 50×50 distance matrices, and $r = 10$ in Figure 5. Each colored line corresponds to an input distance matrix pair. We see that after around 100 random projections, the estimate deviates by less than 10% from the high-fidelity Monte Carlo estimate obtained with 20 000 projections.

Next, we use the same random matrices to display the impact of the quadrature grid r on SFTLB. Here, it is important to notice that the choice of r impacts the ground truth value of SFTLB—notably, a higher choice of r leads to smaller values. This is illustrated for 10 random matrix pairs in Figure 6, where we employ $L = 1000$. Again, each colored line corresponds to a different quantile discretization for a given input distance matrix pair.

Interplay of r, L and the dimension: In this paragraph, we aim to provide additional insights into the dependence of r and L beyond the ablation study. For simplicity, we focus on the classical GW distance without any fused features.

From the theoretical point of view, it is important to note that the quadrature parameter r corresponds to the dimension of $\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}} \in \mathcal{P}(\mathbb{R}^r)$. Generally, slicing techniques are not always well-suited to high-dimensional settings since the required number of random projections can explode. In particular, the Monte Carlo estimate for the sliced Wasserstein distance has a convergence rate of the form

$$\sqrt{\frac{1}{L}}\, C_{\mathrm{MC}}(\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}}) \quad \text{with} \quad C_{\mathrm{MC}}(\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}}) \coloneqq \mathrm{Var}_\theta^{1/2}\left[\mathrm{W}_2^2(\pi_{\theta,\sharp}\,\xi^{\mathcal{Q}},\, \pi_{\theta,\sharp}\,\upsilon_\alpha^{\mathcal{Q}})\right],$$

where Var_θ denotes the variance with respect to uniform spherical projections $\theta \sim \mathcal{U}(\mathbb{S}^{r-1})$, see (Nadjahi et al. 2020). Given arbitary measures on $\mathcal{P}(\mathbb{R}^r)$, we would expect the constant $C_{\mathrm{MC}}(\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}})$ to balloon for high dimensions r. Practically, we can circumvent this issue by choosing a small $r \ll n, m$ as in our experiments, e.g., $r = 10$ or $r = 50$.

However, even beyond this numerical workaround, we observe good classification results for $L \ll r$ in Table 5. We hypothesize that this the effect occurs because $\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}}$ tend to concentrate on a 'smaller' space, especially for large r. Particularly, $\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}}$ are only supported on a cone $S_r = \{0 \leq x_1 \leq \ldots \leq x_r\}$. Given fixed mm-spaces $\mathbb{X}, \mathbb{Y}$, the constant $C_{\mathrm{MC}}(\xi^{\mathcal{Q}}, \nu^{\mathcal{Q}})$ does hence not necessarily become larger for larger r.

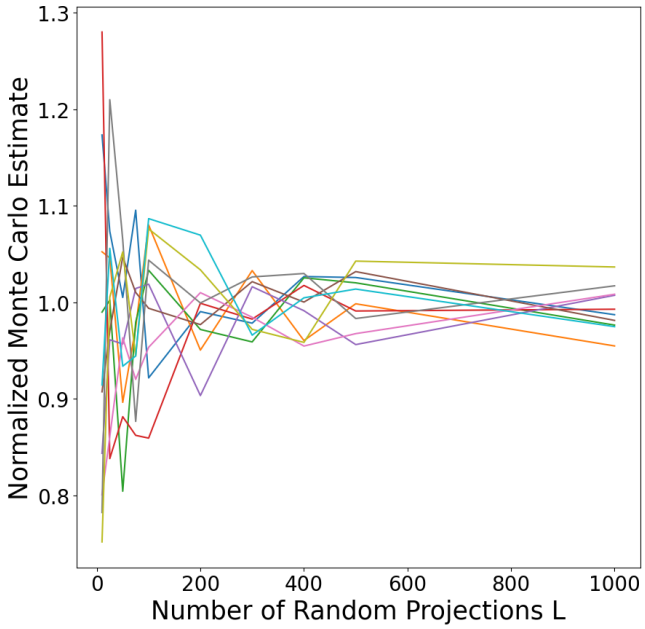


Figure 5: Impact of projection number L on deviation of Monte Carlo SFTLB Estimate. Each line corresponds to multiple SFTLB estimates for a fixed simulated measure pair.

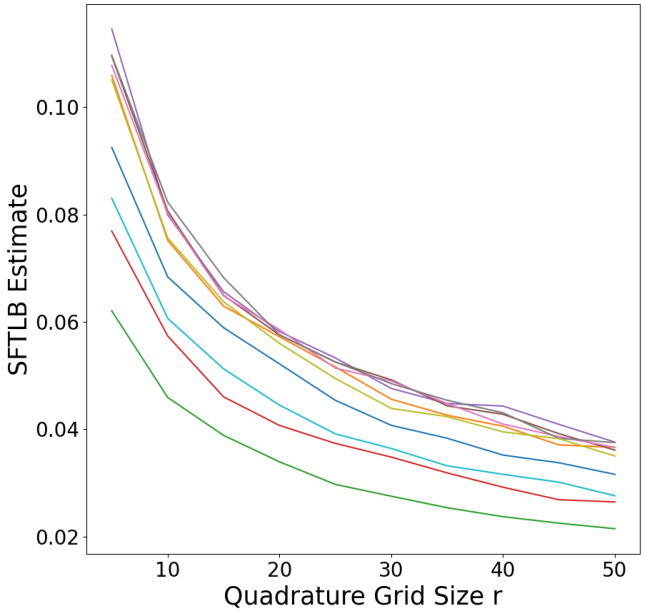


Figure 6: Large quadrature sizes r lead to smaller SFTLB values for the same inputs. Each line corresponds to multiple SFTLB estimates for a fixed simulated measure pair.

Hyperparameter Ablation Study

We repeat the '2D Shapes' and the 'FAUST-500' experiments presented in Table 2 for varying numbers of quadrature points $r = 5, 25, 100$ and random projections $L = 10, 100, 500$ and present the results in Table 5. Higher choices of r and L increase accuracy slightly, but also decrease runtime. As for the hyperparameter α controlling the feature importance, we remark that its influence and optimal value depend on the application. It can be chosen based on domain knowledge or cross-validation strategies, see (Vayer et al. 2020a).

STLB Parameters	2D shapes		FAUST-500	
	acc. (%)	time (ms)	acc. (%)	time (ms)
$r = 5$ & $L = 10$	98.3±2.1	0.8	33.9±5.1	4.7
$r = 5$ & $L = 100$	97.9±2.2	1.2	34.4±5.4	9.4
$r = 5$ & $L = 500$	98.1±2.1	2.9	34.4±5.4	17.1
$r = 25$ & $L = 10$	98.1±1.4	0.6	35.3±5.4	4.6
$r = 25$ & $L = 100$	99.5±1.1	1.5	35.1±5.4	9.9
$r = 25$ & $L = 500$	99.7±0.8	4.0	35.4±5.5	17.6
$r = 100$ & $L = 10$	98.8±1.5	1.2	33.2±5.0	5.1
$r = 100$ & $L = 100$	99.7±0.9	2.0	35.6±5.4	9.9
$r = 100$ & $L = 500$	99.6±0.9	5.1	35.1±5.4	17.7

Table 5: Ablation of KNN Shape classification: Mean accuracy (acc.) with standard deviation and mean runtime (time).

Implementation Details for LGW and qGW

We employ the POT implementation for qGW (Chowdhury, Miller, and Needham 2021), where we set the parameters $n_1 = n_2 = 25$. Moreover, we employ the official GitHub implementation (https://github.com/Gorgotha/LGW/tree/master) for LGW (Beier, Beinert, and Steidl 2022). Here, we calculate the reference space via a GW barycenter of size 50 regarding all inputs for each experiment. Note that the runtime of this linearized method behaves differently from other methods since distance calculations become almost instantaneous after the computation of the reference space and the subsequent alignment of the input data. However, this preprocessing is very time-consuming and most of the time is spent on the barycenter calculation. Hence, we present the overall run time divided by the number of distances in Table 2.

We tried to choose suitable hyperparameters for both methods. Note that this choice involves a trade-off between quality and run time. As a result, the presented methods can achieve better classification results at the cost of a longer runtime. In particular, the reference space of our LGW implementation relies on a barycenter computation based on all inputs, whereas LGW allows accelerated computations by using a subsampled barycenter (at the cost of potential performance losses).

Lastly, we note that qGW offers the advantage of GW transport plans, and the LGW framework has the unique benefit of enabling highly efficient distance calculations for new and unseen data.

Additional Large-Scale Experiment

Adding to the experiments in Table 2, we conduct an additional shape classification experiment on a larger dataset. In particular, we focus on the test split of the ModelNet-10 dataset (Uy et al. 2019). This dataset split consists of 908 shapes from 10 different classes. We randomly downsample each shape to an Euclidean point cloud of size 50.

Subsequently, we employ the same classification pipeline as in the other experiments presented in Table 2. As a result, we get the following classification accuracies: 26.4% $\pm$ 1.6% for SLB, 27.9% $\pm$ 1.7% for TLB, 27.3% $\pm$ 1.7% for AE, 26.8% $\pm$ 1.7% for STLB and 27.3% $\pm$ 1.7% for GW. Again, STLB performs on par with other approaches.